

Computer Graphics

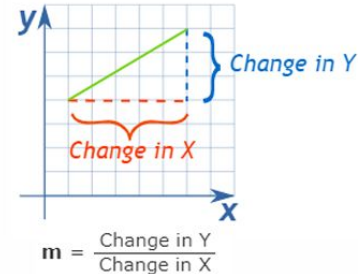
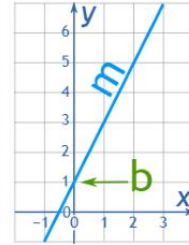
Chapter 3 : Line Drawing Algorithms

0xSaad Chapters

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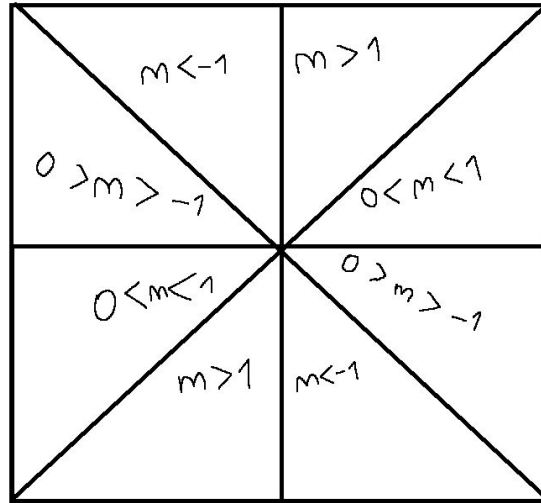
The equation of a straight line is $y = mx + b$

m is the slope or gradient



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Slope : dy / dx , where $dy = y_2 - y_1$ and $dx = x_2 - x_1$



$$m = \frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1}$$

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DDA or Digital Differential Analyzer : Simple to learn and easy algorithm using float numbers

Using slope and steps to increments x and y

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Slope = dy / dx

Steps = $\max(|dx|, |dy|)$

Inc_x = $dx / steps$

Inc_y = $dy / steps$

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DDA Example :

DDA

من النقطة (1,1) إلى النقطة (4,6)

$$\begin{matrix} x_1 & y_1 & & x_2 & y_2 \\ (1,1) & + & (4,6) \end{matrix}$$
$$dx = x_2 - x_1 = 4 - 1 = 3$$
$$dy = y_2 - y_1 = 6 - 1 = 5$$
$$\text{Steps} = \max(|dx|, |dy|) = 5$$
$$\text{inc}_x = \frac{dx}{\text{Steps}} = \frac{3}{5}$$
$$\text{inc}_y = \frac{dy}{\text{Steps}} = \frac{5}{5} = 1$$

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DDA Example : the table :

Step	Point	Rule
0	(1,1)	$X = 1+0.6, Y=1+1$
1	(1.6,2)	$X=1.6+0.6, Y=2+1$
2	(2.2,3)	$X=2.2+0.6, Y=3+1$
3	(2.8,4)	$X=2.8+0.6, Y=4+1$
4	(3.4,5)	$X=3.4+0.6, Y=5+1$
5	(4,6)	—

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Bresenham Algorithm : when $m > 1$ we will increment $y+1$ in any step and x was incremented by “ p ” or called “decision parameter”

When $p \geq 0$ we will increase x

Put when $p < 0$ we will increment y by 1

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Bresenham Algorithm : when $m < 1$ we will increment $x+1$ in any step and y was incremented by p

When $p \geq 0$ we will increase y
Put when $p < 0$ we will increment x by 1

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Bresenham Algorithm :

$$m = \frac{dy}{dx}$$

$$m > 1$$

Y++

X by P

$$P_0 = 2dx - dy$$

$$P \geq 0 = P + 2dx - 2dy$$

$$P < 0 = P + 2dx$$

$$m < 1$$

X++

Y by P

$$P_0 = 2dy - dx$$

$$P \geq 0 = P + 2dy - 2dx$$

$$P < 0 = P + 2dy$$

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Bresenham Algorithm Example:

من النقطة (1,1) إلى النقطة (7,4)

$dx = 7 - 1 = 6$
 $dy = 4 - 1 = 3$
 $m = \frac{dy}{dx} = \frac{3}{6} = 0.5$
 $m < 1$
 $2dy = 6$
 $2dx = 12$
 $p_0 = 2dy - dx = 6 - 12 = -6$

1,1 → 4,7

$p \geq 0 \Rightarrow p = p + 2dx - 2dy$
 $p < 0 \Rightarrow p = p + 2dx$

Step	Point	P	Operation	Decision
0	(1,1)	-6	$-6 + 6 = 0$	$y++$
1	(1,2)	0	$0 + 6 - 12 = -6$	$y++, x++$
2	(2,3)	-6	$-6 + 6 = 0$	$y++$
3	(2,4)	0	$0 + 6 - 12 = -6$	$y++, x++$
4	(3,5)	-6	$-6 + 6 = 0$	$y++$
5	(3,6)	0	$0 + 6 - 12 = -6$	$y++, x++$
6	(4,7)	—	—	—

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DDA vs Bresenham

Advantages

DDA	Bresenham
1. Easy to learn 2. Simple to Implement	1. Faster than DDA 2. Using integer numbers
1. Slower than Bresenham 2. Not Suitable for hardware	3. Using Only for Basic Line Drawing

Disadvantages